Module 4

Module 4: Linear Classification

Spring 2026

Where We Are in the Course?

So far we have focused on:

- Estimating continuous parameters
- Minimizing χ^2 or maximizing likelihood
- Interpreting uncertainty from curvature

Where We Are in the Course?

So far we have focused on:

- Estimating continuous parameters
- Minimizing χ^2 or maximizing likelihood
- Interpreting uncertainty from curvature

Now we switch problems:

Not: What value best describes the data? → But: <u>Which hypothesis produced the data?</u>

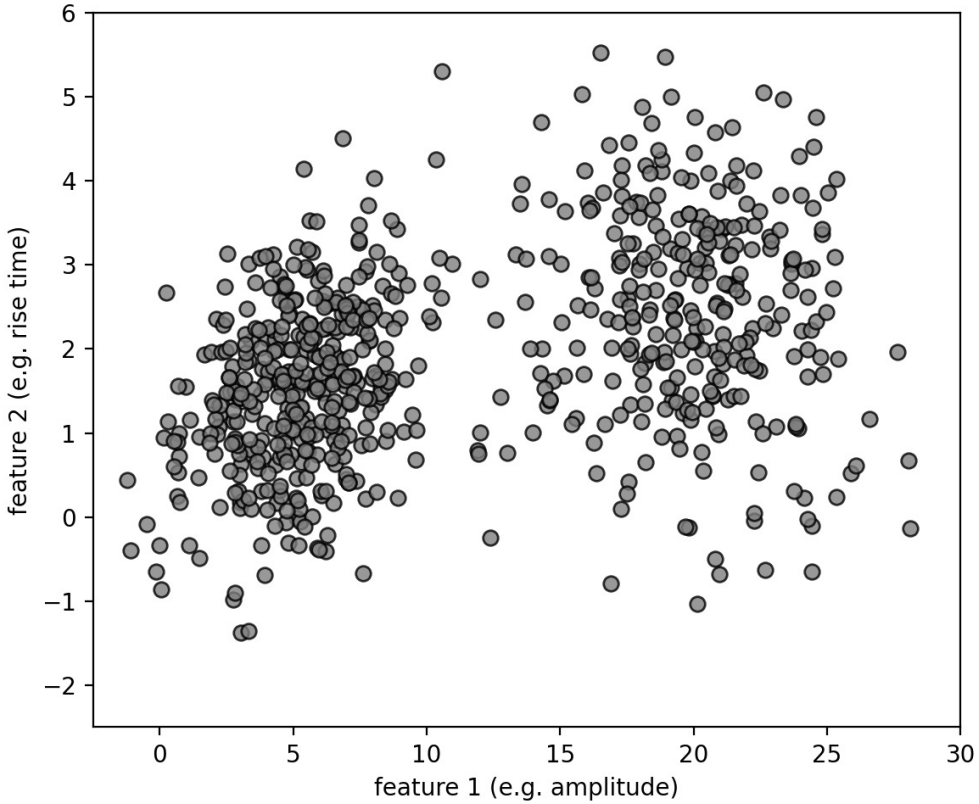
This is a **classification problem.**

Examples of Classification in Physics

- Signal vs background event
- Good pulse vs noise trigger
- Electron recoil vs nuclear recoil
- Gamma vs neutron

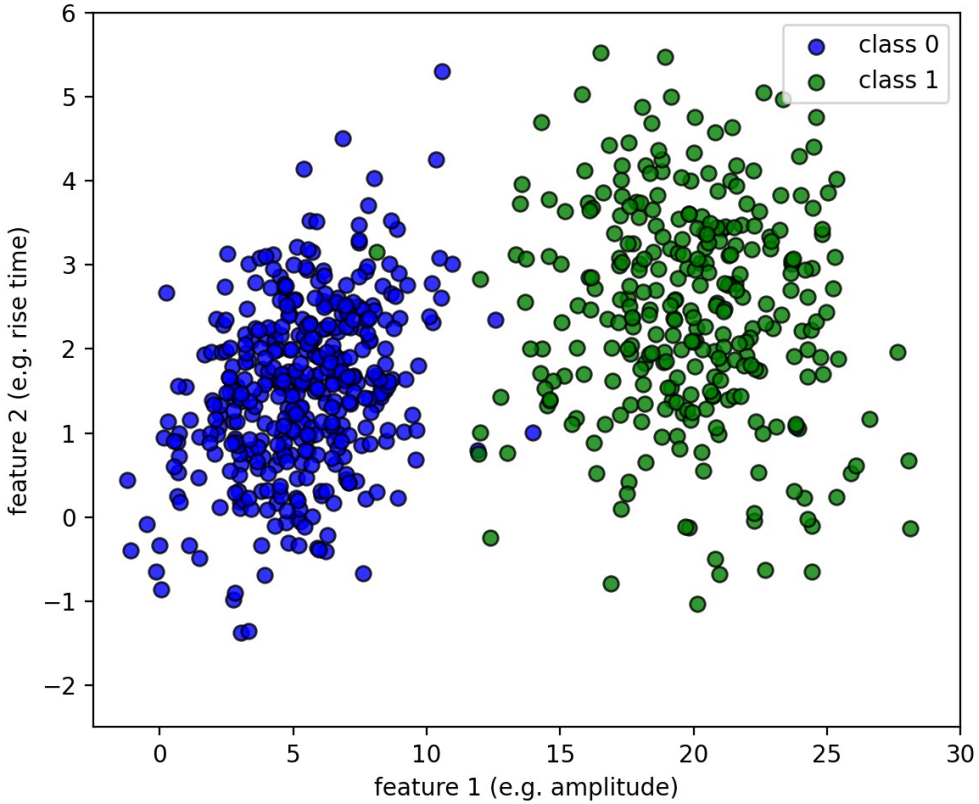
Instead of estimating a number, we assign a label:

$$y \in \{0, 1\}$$



Examples of Classification in Physics

- Signal vs background event
- Good pulse vs noise trigger
- Electron recoil vs nuclear recoil
- Gamma vs neutron



Instead of estimating a number, we assign a label:

$$y \in \{0, 1\}$$

Goal:

Predict class from measured features

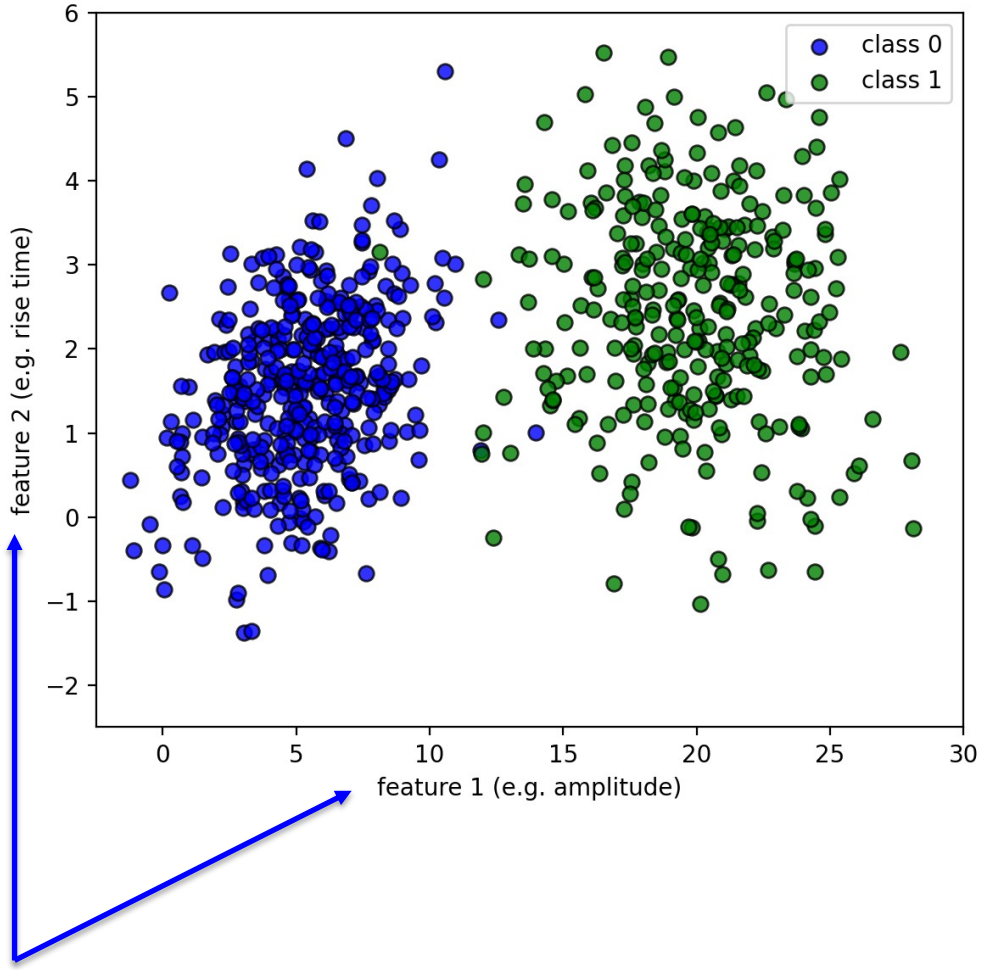
Feature Space

Each event is described by measurements:

$$\mathbf{x} = (x_1, x_2, x_3, \dots)$$

Examples:

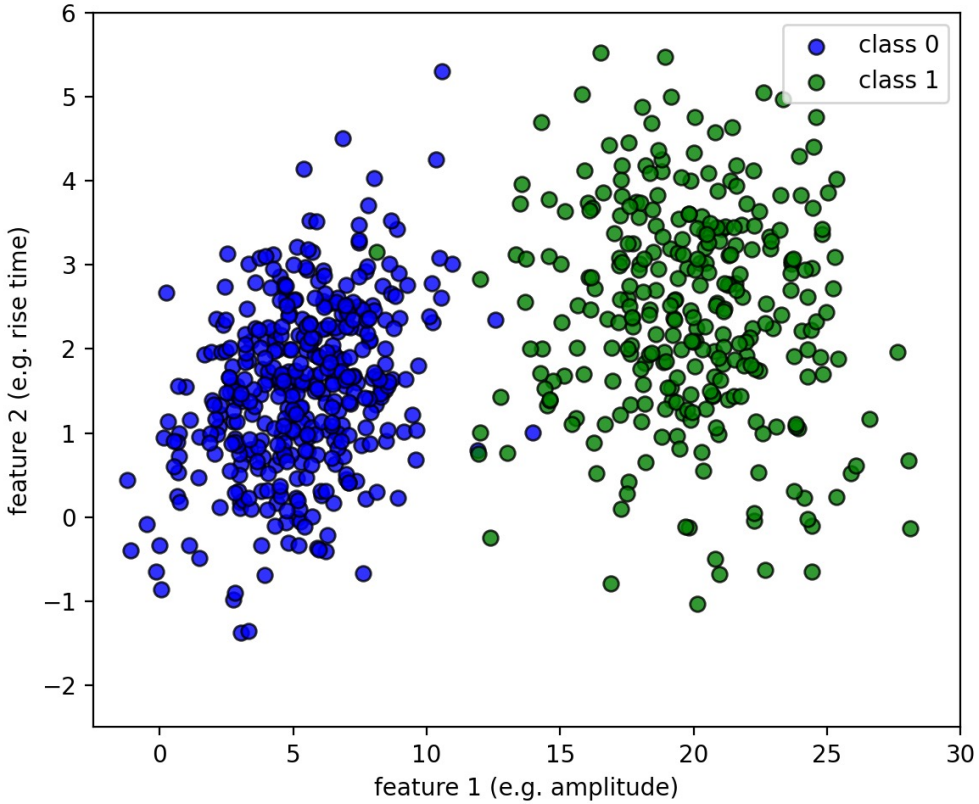
- pulse height
- rise time
- pulse integral
- frequency content



An event becomes a point in **feature space**.

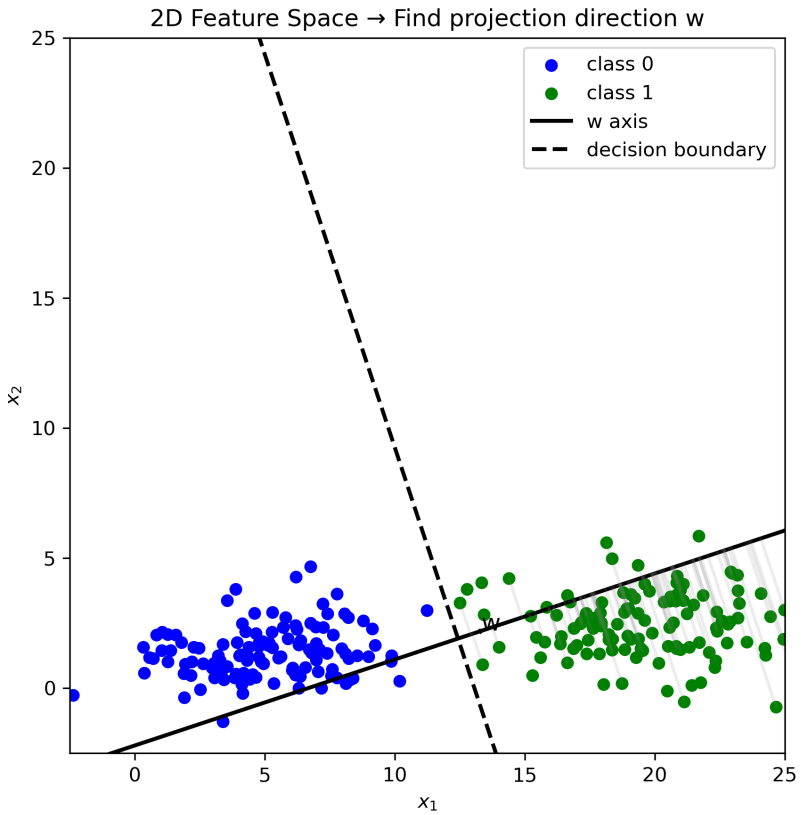
Linear Decision Rule

- No single cut we can apply in single feature



Linear Decision Rule

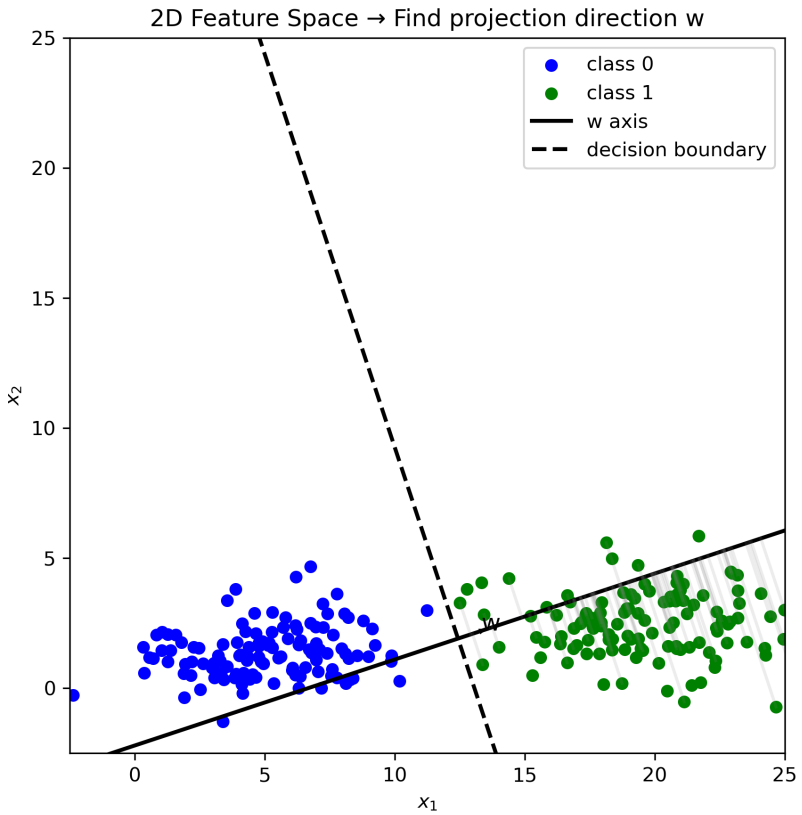
- No single cut we can apply in single feature
- Instead, we choose a direction w and look along it



Linear Decision Rule

- No single cut we can apply in single feature
- Instead, we choose a direction w and look along it
- We classify using a linear function:

$$g(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b$$

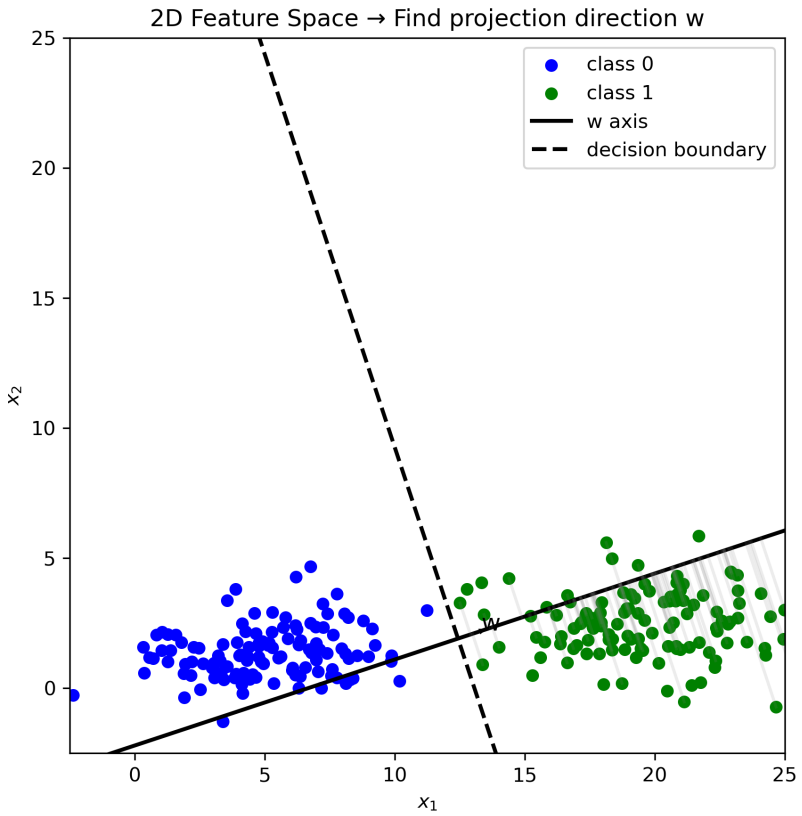


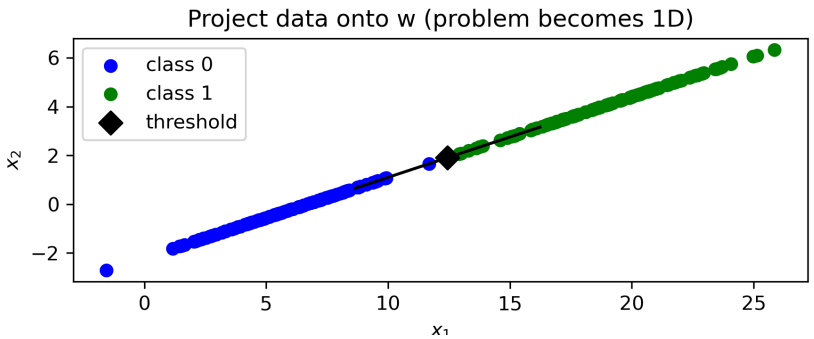
Linear Decision Rule

- No single cut we can apply in single feature
- Instead, we choose a direction w and look along it
- We classify using a linear function:

$$g(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b$$

- Classification reduces to 1D after projection



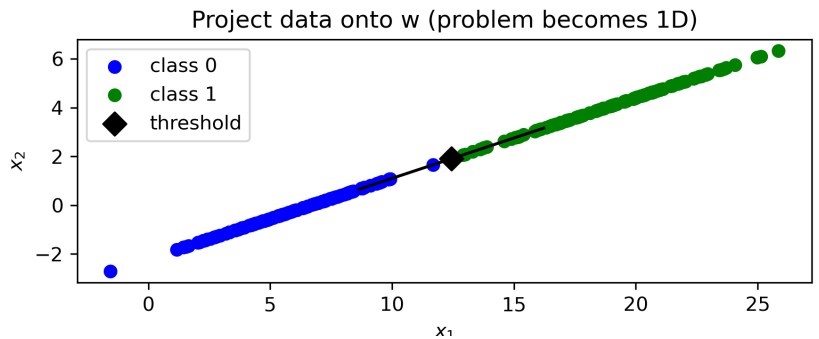


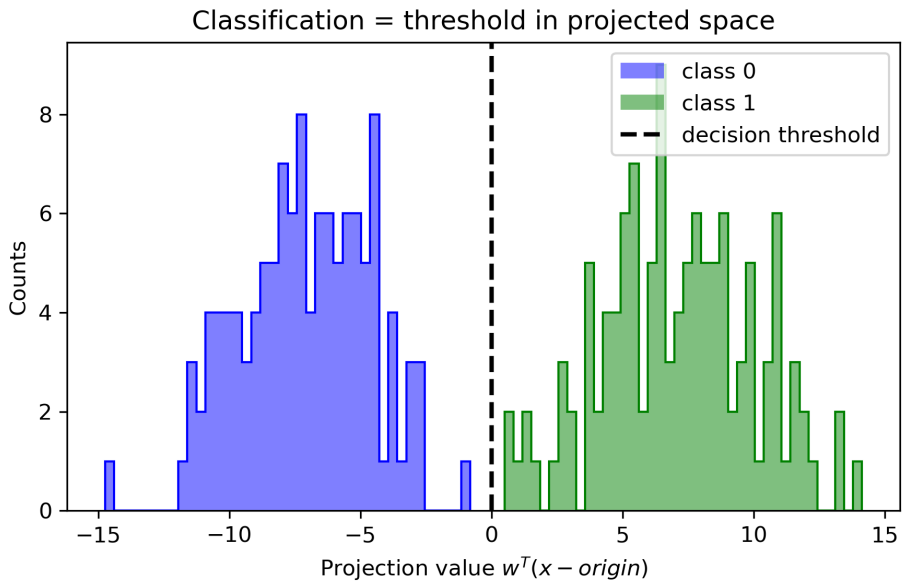
Linear Decision Rule

- No single cut we can apply in single feature
- Instead, we choose a direction w and look along it
- We classify using a linear function:

$$g(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b$$

- Classification reduces to 1D after projection





$$\hat{y} = \begin{cases} 1 & g(\mathbf{x}) > 0 \\ 0 & g(\mathbf{x}) < 0 \end{cases}$$

Linear Decision Rule

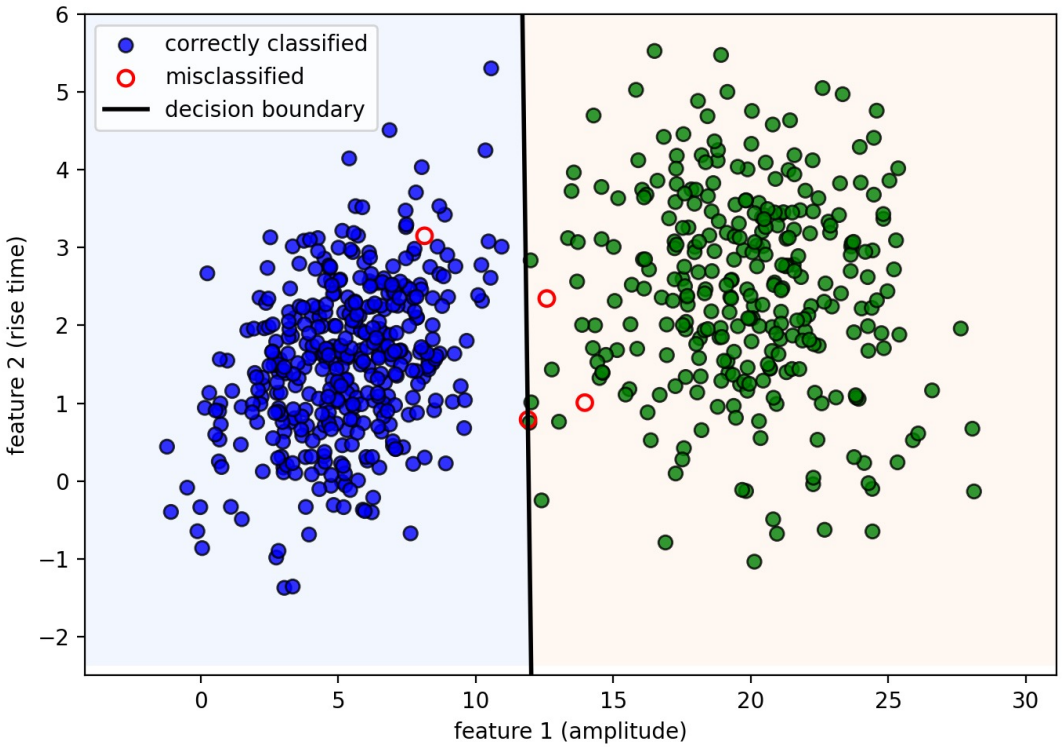
We classify using a linear function: $$g(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b$$

Decision rule: $$\hat{y} = \begin{cases} 1 & g(\mathbf{x}) > 0 \\ 0 & g(\mathbf{x}) < 0 \end{cases}$$

The equation:

$$\mathbf{w}^T\mathbf{x} + b = 0$$

defines a **decision boundary**



Geometry

- In 2D: line
- In 3D: plane
- In higher dimensions: hyperplane

The classifier separates feature space into two regions

Classification is geometry

We are fitting a boundary, not a curve.

Connection to Linear Regression

Regression model:

$$y = \mathbf{w}^T \mathbf{x} + b$$

Classification model:

$$y = \mathrm{sign}(\mathbf{w}^T \mathbf{x} + b)$$

Connection to Linear Regression

Regression model:

$$y = \mathbf{w}^T \mathbf{x} + b$$

Classification model:

$$y = \mathrm{sign}(\mathbf{w}^T \mathbf{x} + b)$$

Same model → Different interpretation → Different loss function

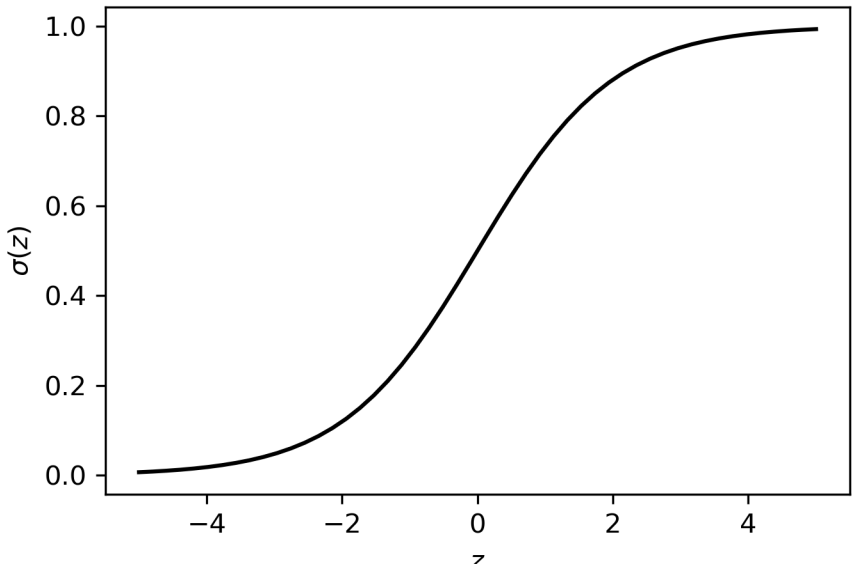
Probabilistic Interpretation

Instead of a hard decision, predict probability:

$$P(y = 1|\mathbf{x})$$

We map the linear function to a probability using:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



$$P(y = 1|\mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x} + b)$$

Linear boundary + squashing function = classifier

Decision Rule

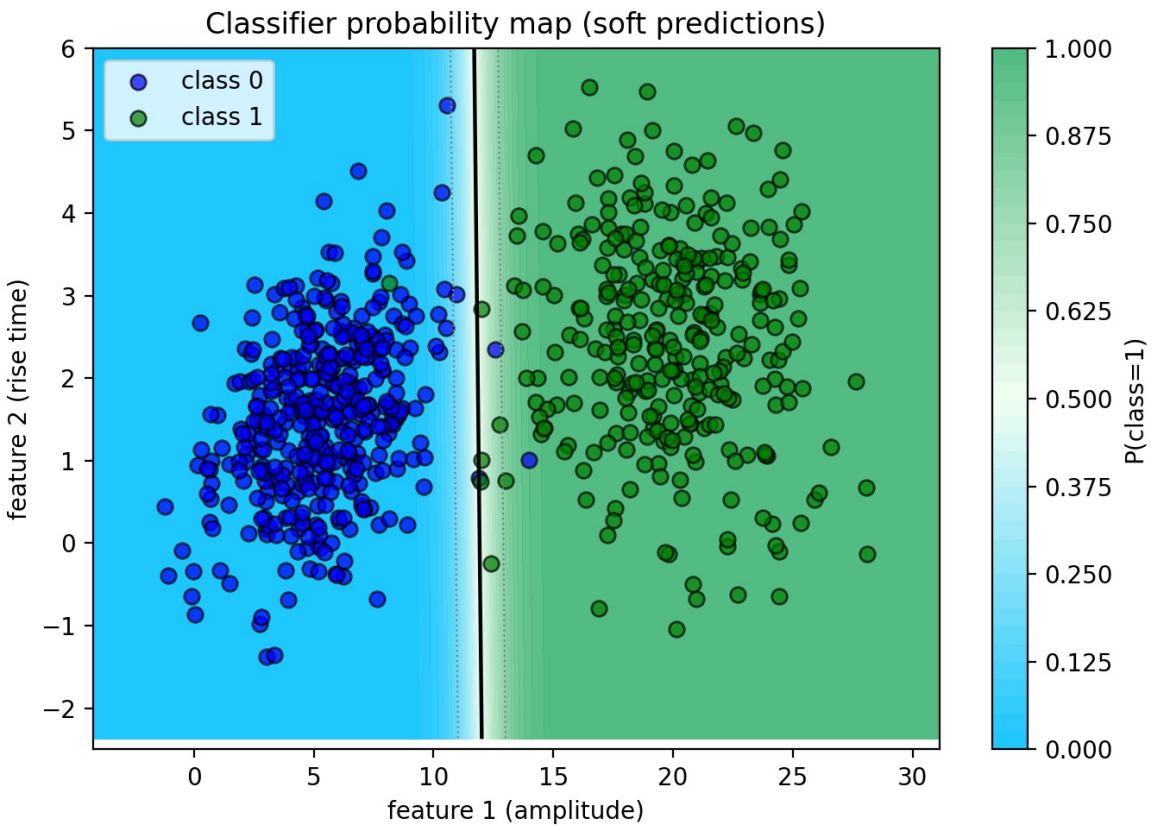
$$P(y = 1|\mathbf{x}) > 0.5 \Rightarrow \text{class } 1$$

$$P(y = 1|\mathbf{x}) < 0.5 \Rightarrow \text{class } 0$$

This is equivalent to:

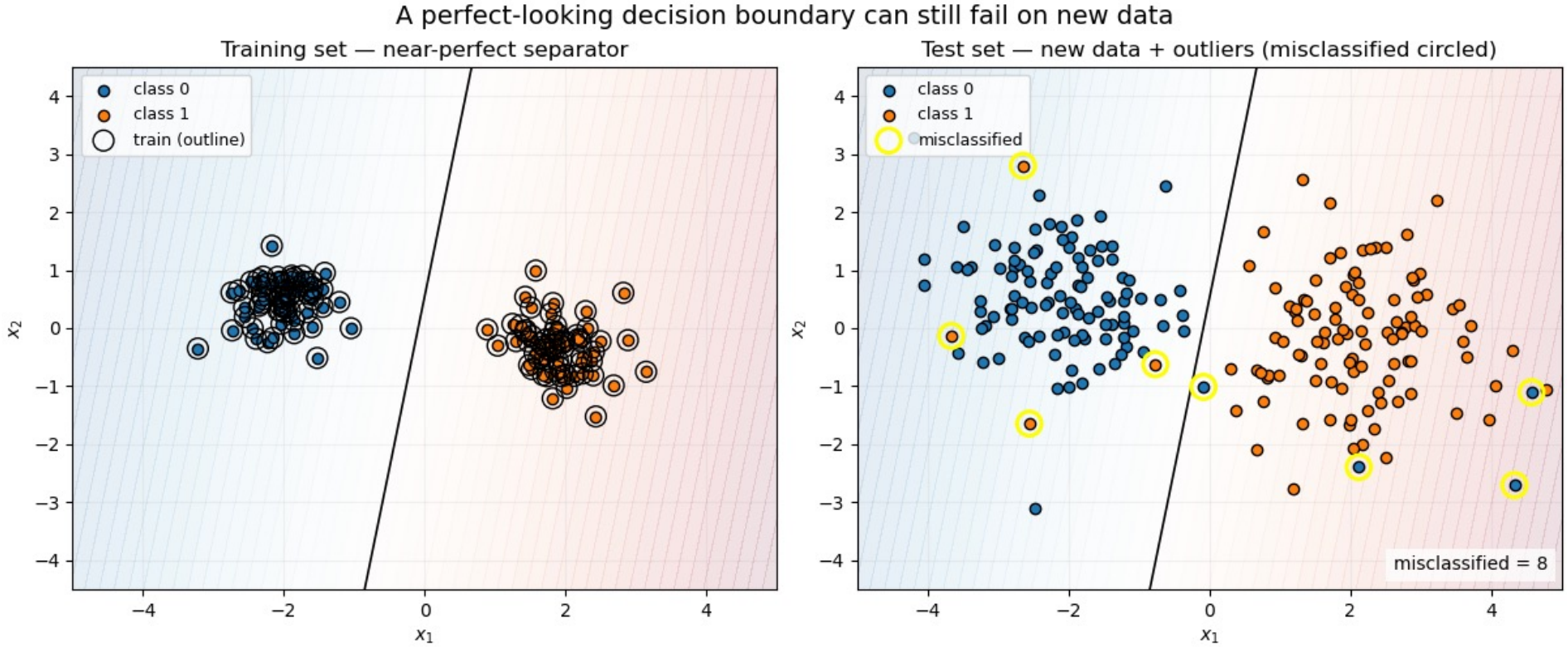
$$\mathbf{w}^T\mathbf{x} + b > 0$$

So probability and geometry give the same boundary.



A perfect boundary can still be wrong

A classifier is not judged by how well it fits the training data, but how well it predicts unseen data



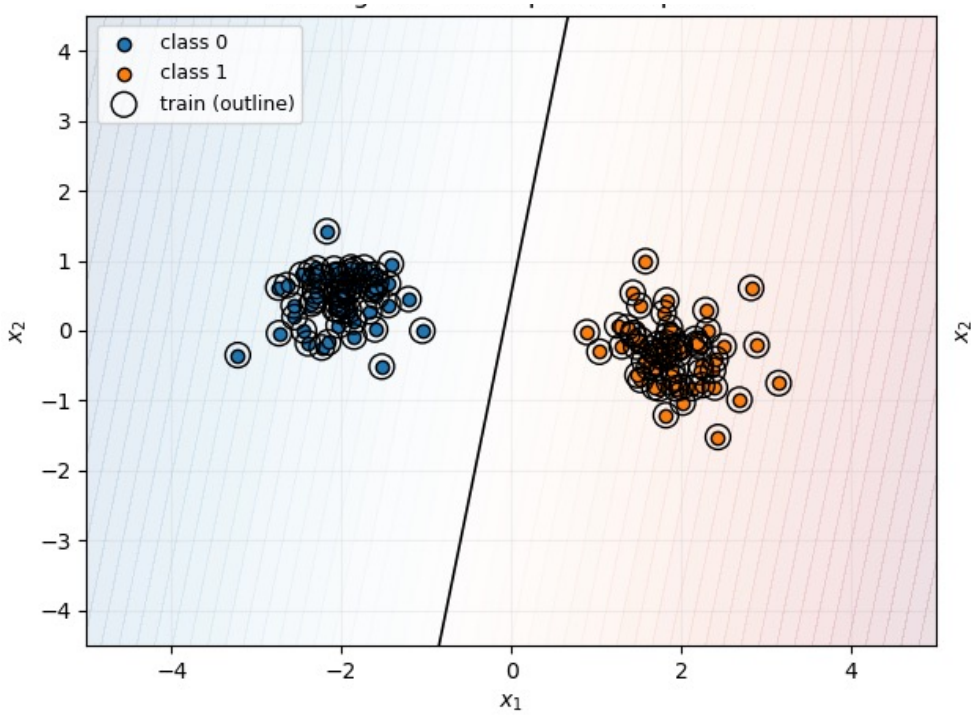
Training error ≠ Prediction error

Train vs Test

1. Split dataset:

$$\text{Data} = \text{Train set} + \text{Test set}$$

2. Train:
   - Learn parameters w, b



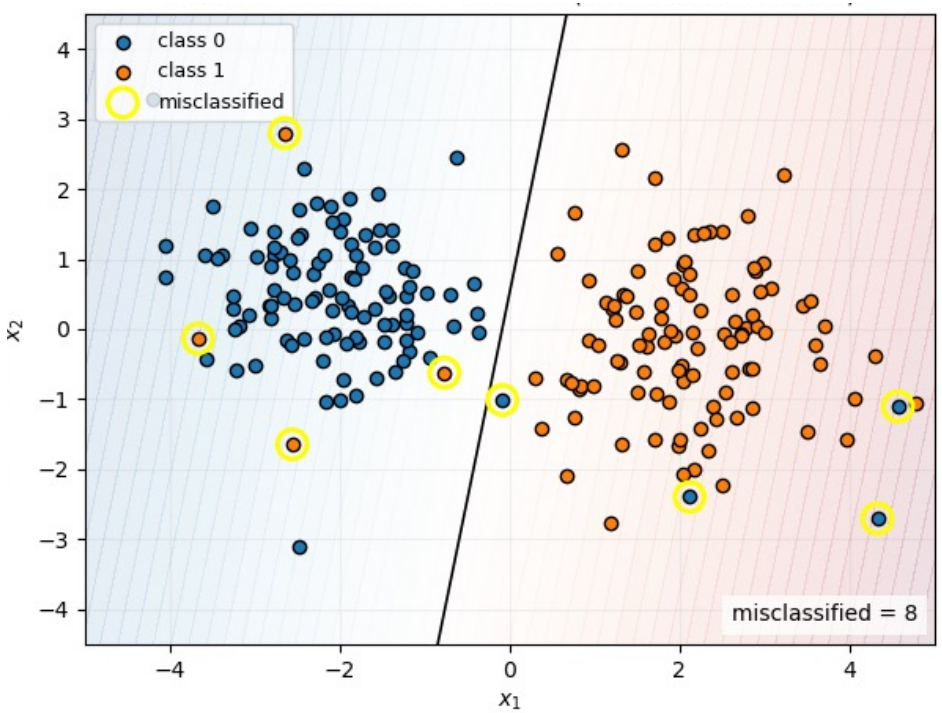
Train vs Test

1. Split dataset:

$$\text{Data} = \text{Train set} + \text{Test set}$$

2. Train:
   - Learn parameters w, b
3. Test:
   - Evaluate performance

The test is a ‘mock future experiment’



The confusion matrix

For binary classification, we can have the following outcomes

	Predicted Positive	**Predicted Negative**
Actually Positive	**TP** = True Positive	**FN** = False Negative
Actually Negative	**FP** = False Positive	**TN** = True Negative

want to maximize the diagonals and minimize the off-diagonals

The threshold problem

Most classifiers output a score:

$$g(x) = w^T x + b$$

We convert score → decision by threshold:

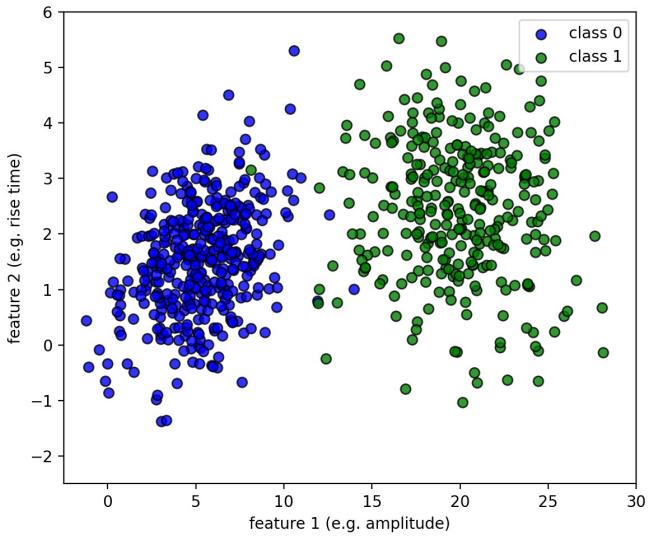
$$g(x) > t \rightarrow \text{signal}$$

Changing t:

- increases efficiency
- increases background

ROC Curve

Example: signal vs background

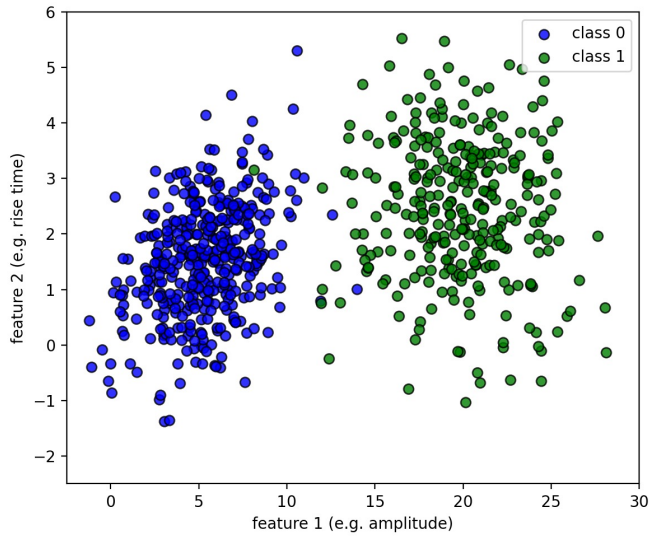


$$\text{Efficiency (TPR)} = \frac{TP}{TP + FN}$$

$$\text{Background leakage (FPR)} = \frac{FP}{FP + TN}$$

ROC Curve

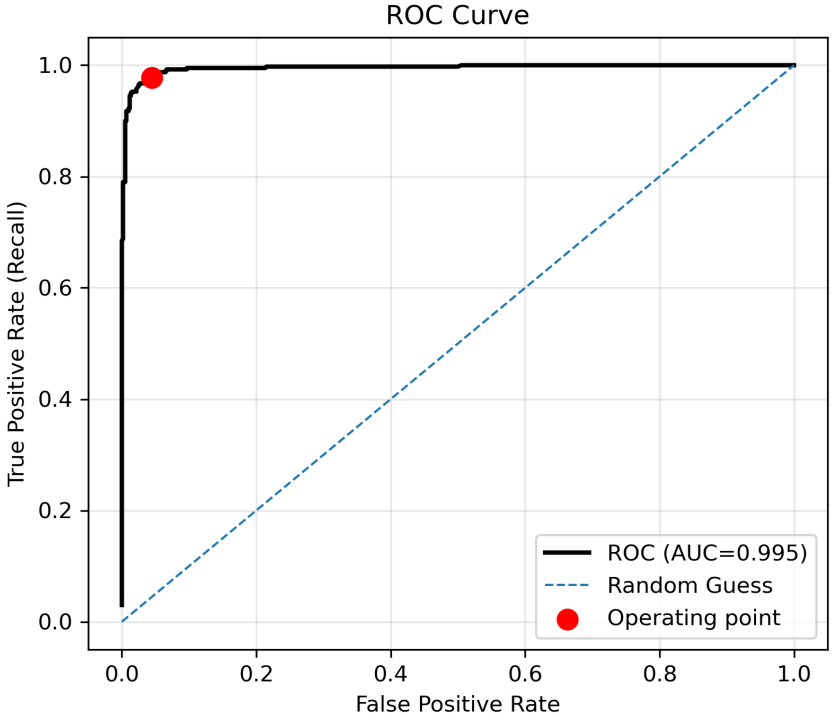
Example: signal vs background

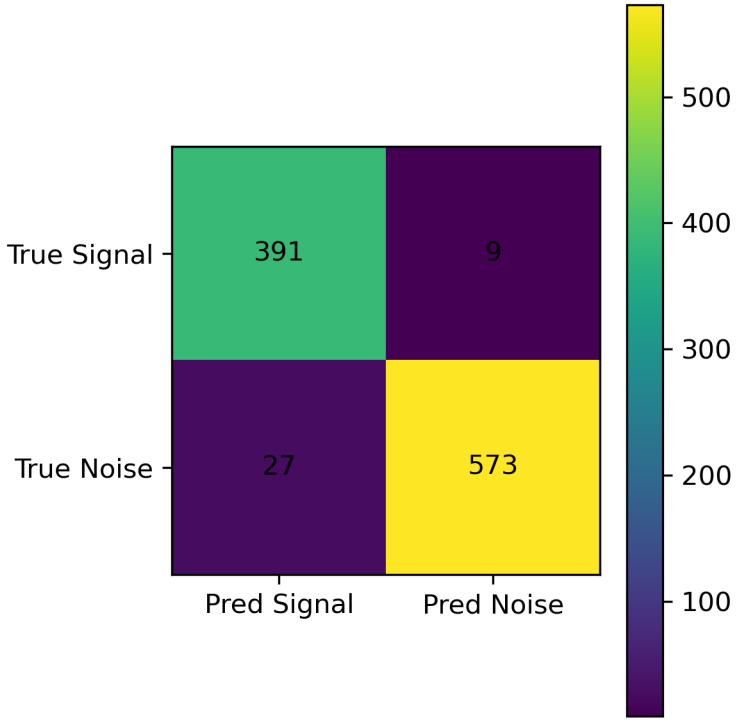


We change our threshold to maximize true positive and minimize false positives

$$\text{Efficiency (TPR)} = \frac{TP}{TP + FN}$$

$$\text{Background leakage (FPR)} = \frac{FP}{FP + TN}$$





Key Takeaways

- Classification predicts discrete labels
- Data live in feature space
- A linear classifier draws a hyperplane
- Logistic function converts score to probability
- Same structure as regression — new interpretation

Next time: implementing and visualizing classifiers